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Presentation on

RAG-Enhanced Transformer Framework for Context-Aware Polysemy Disambiguation and Automatic Title Generation

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Abstract

1. Problem Statement

Polysemous words can have multiple meanings, making it difficult for NLP systems to correctly understand text context and generate suitable titles.

2. Objective

To improve contextual understanding by resolving polysemous words and automatically generating meaningful, context-appropriate titles from textual content.

3. Proposed Solution

A RAG-enhanced Transformer framework is proposed to retrieve relevant external knowledge, disambiguate polysemous words, and generate suitable titles based on the given content.

4. Methodology

The system preprocesses the input text, identifies ambiguous words, retrieves relevant context using RAG, performs contextual analysis using a Transformer model, and generates and evaluates an appropriate title.

5. Expected Outcome

The system is expected to provide better contextual understanding, accurate polysemy disambiguation, and relevant, meaningful titles for diverse textual content.

Keywords: RAG (Retrieval-Augmented Generation), Transformer, NLP (Natural Language Processing), Polysemy, Contextual Understanding, Disambiguation, Automatic Title Generation.

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Introduction

1. Natural Language Processing

Natural Language Processing (NLP) enables **computers to understand and interpret human language**, but understanding the correct meaning of words depends strongly on context.

2. Polysemy and Contextual Ambiguity

Polysemy occurs when a single word has **multiple related meanings**, making accurate interpretation difficult when the available context is limited or ambiguous.

3. Transformer-Based Understanding

Transformer models provide **strong contextual understanding** by analysing relationships between words, but may require additional information when the input does not provide enough context.

4. RAG-Enhanced Context

Retrieval-Augmented Generation (RAG**) retrieves relevant information** from external knowledge sources and provides additional context to help the Transformer disambiguate polysemous words.

5. Automatic Title Generation

The proposed framework **combines RAG and Transformer-based analysis** to understand the complete content, resolve ambiguous meanings, and automatically generate a relevant and meaningful title.

Literature Survey

S.No.	Author(s) & Year	Research / Method Used	Key Findings	Limitations / Research Gap	Proposed Enhancement
1	Bevilacqua et. al. (2021)	Word Sense Disambiguation (WSD) Survey	Reviewed recent advances and representation-learning improvements.	WSD still depends on available sense inventories and contextual resources.	Use external retrieved knowledge to provide richer context.
2	Li et al. (2024)	Transformer + BiLSTM for WSD	Combined Transformer with BiLSTM for Chinee WSD	Focused mainly on Chinese and specific datasets.	Extend disambiguation across diverse domains.
3	Abeysiriwardana & Sumanathilaka (2024)	Lexical Ambiguity Detection and WSD	Reviewed modern approaches for resolving ambiguity.	Scarcity of sense-annotated data and complex texts.	Integrate RAG and broader knowledge sources.
4	Yae et al. (2024/2025)	Large Language Models for WSD	Evaluated LLMs for word sense identification.	LLM performance varies and struggles with ambiguities.	Combine retrieval with LLM reasoning.
5	Zhang et al. (2025)	RoDEval – LLM-based WSD Evaluation	Evaluated LLMs' robustness and disambiguation	LLMs may lack sense coverage and show bias.	Enhance WSD with reliable external knowledge.

Software Requirements	Hardware Requirements
Operating System: Windows	**Processor:** Intel Core i5
Programming Language: Python	**RAM:** Minimum 8 GB
Development Environment: VS Code	**Storage:** Minimum 20 GB available space
NLP & Transformer: Hugging Face Transformers	**GPU:** NVIDIA GPU with at least 4 GB VRAM
RAG Framework: Lang Chain / LlamaIndex	**Internet Connection:** Required for external knowledge retrieval and development resources
Libraries: PyTorch, Pandas, NumPy	**Input/Output Devices:** Keyboard, Mouse and Standard Monitor